AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 2)

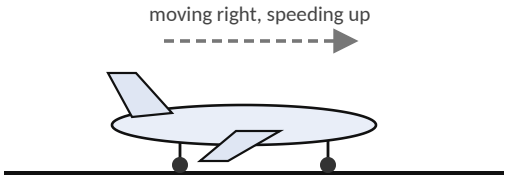
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

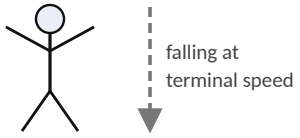
For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) For each situation, draw and label the forces (not components) exerted on the object named. Draw each force as its own arrow starting on the dot and pointing away from it, with lengths that show which forces are larger.



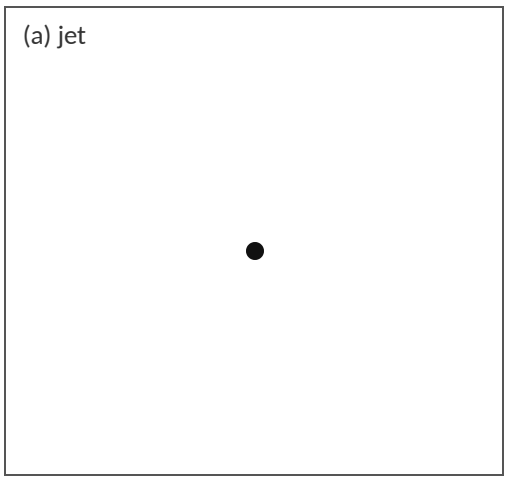
(a)

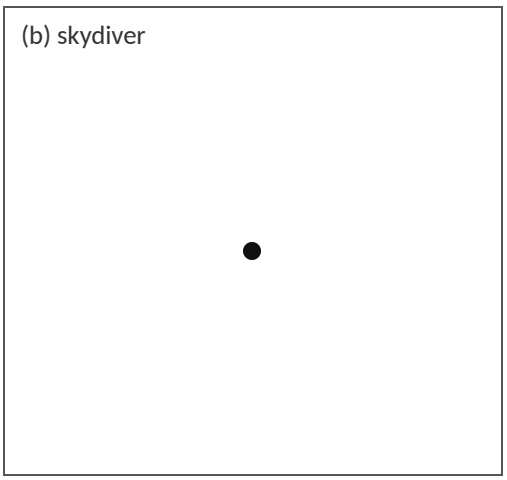


(b)

(a) A jet moves to the right along a level runway and is speeding up. It has not lifted off yet: its wheels are still on the runway, and the wings already produce some lift. Friction on the wheels is negligible. Draw the free-body diagram of the jet.

(b) A skydiver falls straight down at terminal speed. Draw the free-body diagram of the skydiver.





(c) In part (a), is the normal force from the runway on the jet less than, equal to, or greater than the jet's weight? Explain.

(d) A student says: "The skydiver in part (b) is moving down, so there must be a net force on her pointing down." Do you agree or disagree? Explain using Newton's laws.

2. (25 points) A rocket is launched straight up from rest. At liftoff its total mass, including fuel, is M = 3.40×10^4 kg. Its engines push it up with a constant thrust force of 5.10×10^5 N. As the engines burn fuel, the rocket's mass decreases. Air resistance is negligible, and g = 9.8 m/s² the whole time.



(a) Draw the free-body diagram of the rocket just as it lifts off. Then calculate the rocket's acceleration at liftoff, magnitude and direction.

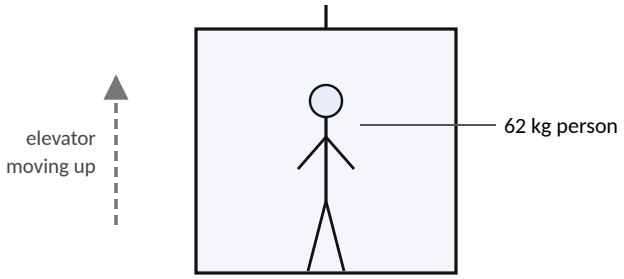
(a) rocket, at liftoff

(b) Later, the rocket is still moving straight up with the same thrust, and its acceleration is 7.2 m/s² upward. Calculate the rocket's mass at that moment. Then calculate the mass of fuel it has burned since liftoff.

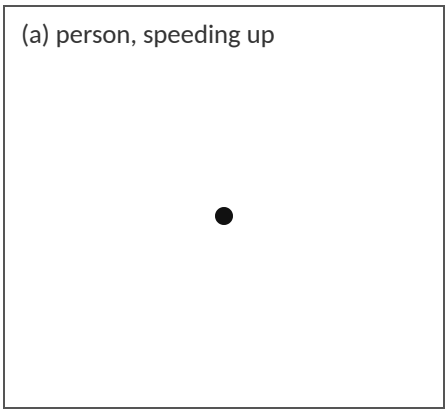
(c) The thrust never changes, but the rocket's acceleration keeps increasing as it rises. Explain why, using Newton's second law.

(d) The thrust is the force of the exhaust gases on the rocket. Identify its Newton's third law partner: what exerts it, on what object, its magnitude, and its direction.

3. (25 points) A 62 kg person stands on the floor of an elevator that is moving up.



(a) The elevator is speeding up at 1.4 m/s². Draw the free-body diagram of the person. Then calculate the normal force from the floor on the person.

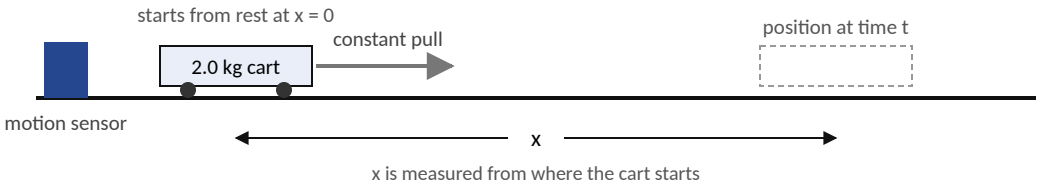


(b) Later, the elevator moves up at a constant speed. Calculate the normal force on the person.

(c) Near the top floor, the elevator is still moving up but is slowing down at 2.1 m/s². Calculate the normal force on the person.

(d) In part (a), what force does the person exert on the floor? Give its magnitude and direction, and name the law you used.

4. (25 points) A 2.0 kg cart starts from rest on a level track. A string pulls it with a constant net force. A motion sensor measures the cart's position x at several times t, with x measured from where the cart starts. Because the cart starts from rest at x = 0 and its acceleration is constant, $x = \frac{1}{2}at^2$.

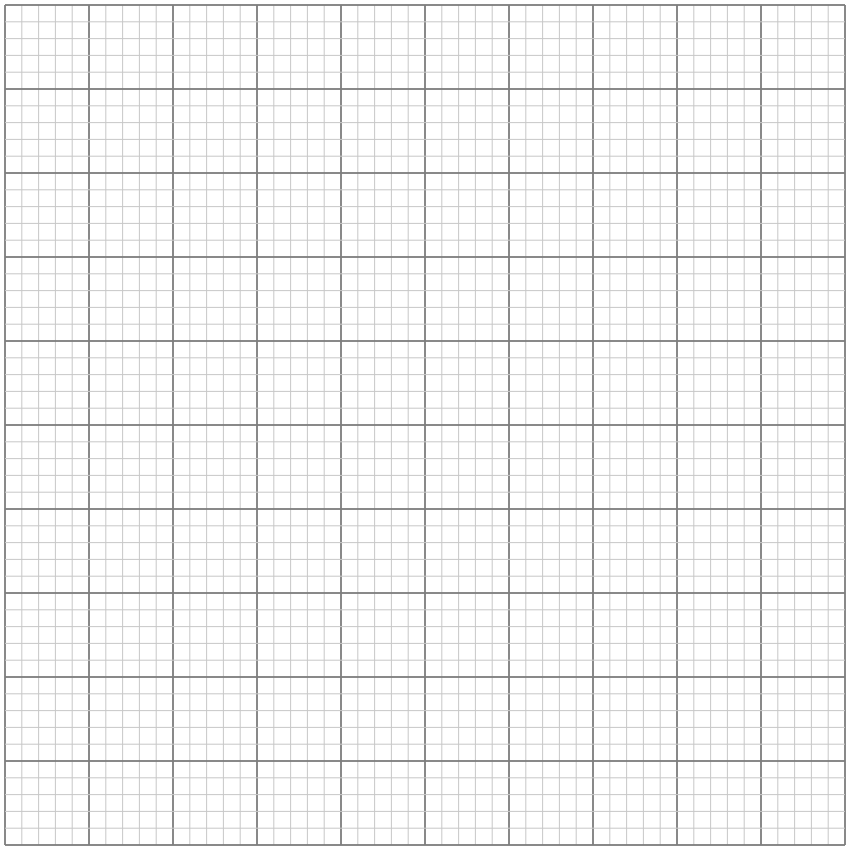


Time t (s)	Position x (m)	t^2 (s^2)
0.50	0.10	
1.00	0.39	
1.50	0.92	
2.00	1.57	
2.50	2.54	
3.00	3.56	

(a) Calculate t^2 for each trial and write it in the blank column. Show one calculation below with units.

You will graph x on the vertical axis and t^2 on the horizontal axis on the next page.

(b) Plot x (vertical axis) against t^2 (horizontal axis) on the grid. Label each axis with the quantity and units, and choose a scale that uses more than half the grid.



(c) Draw the best-fit line. Calculate its slope using two points on the line that are far apart. Show the two points and include units.

(d) Starting from $x = \frac{1}{2}at^2$, state what the slope of your line represents. Then use your slope to calculate the cart's acceleration and the net force on the cart.

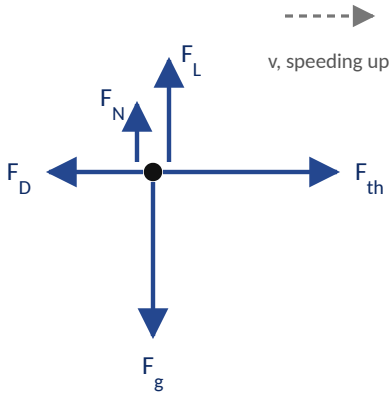
Answer Key – Practice Test, Version 2 – Introductory Dynamics

Worked step by step, the same way as the class notes: sketch, free-body diagram, $a = F_{net}/m$, forces with signs, algebra first, then numbers with units. $g = 9.8\ m/s^2$; answers match the significant figures of the givens.

Every problem: $a = \Sigma F/m$ (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs ($F_{th} - F_D$, $F_N + F_L - F_g$, $F_{th} - F_g$, $F_N - F_g$), swap $F_g = mg$, solve in letters, then plug in.
Kinematics, the first three on the sheet: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagrams (25 points)

(a) jet on the runway



Step 1 **List every force on the jet.**

Gravity F_g pulls down. The runway pushes up on the wheels (F_N). The air pushes up on the wings (F_L). The engines push the jet forward (F_{th}) and the air pushes back (F_D). Wheel friction is negligible, so there is no F_f.

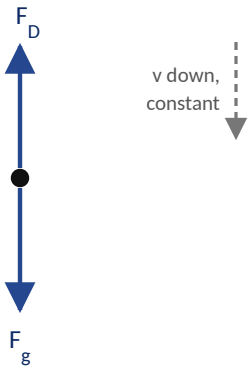
Step 2 **Check the lengths.**

Horizontal: speeding up to the right, so a and F_{net} point right: $F_{th} > F_D$. The F_{th} arrow is longer.

Vertical: no vertical acceleration on the runway, so $F_N + F_L = F_g$. The two up arrows together are as long as F_g.

Watch out: F_N and F_L come from two different objects (the runway and the air), so they are two separate arrows.

(b) skydiver



Step 3 **(b) Terminal speed is a constant velocity.**

Constant velocity means a = 0, so $F_{net} = 0$: drag F_D up balances gravity F_g down. $F_D = F_g$, equal arrows.

Watch out: no extra downward arrow for the motion. Moving down does not need a downward net force.

Step 4 **(c) Normal force on the jet compared with its weight.**

Vertical: $a = \frac{F_{net}}{m} = 0 \rightarrow F_N + F_L - F_g = 0 \rightarrow F_N = F_g - F_L$

Less than the weight The lift carries part of the weight, so the runway only has to push up with $F_g - F_L$.

Step 5 **(d) Disagree.**

Terminal speed is a constant velocity, so a = 0. $a = \frac{F_{net}}{m} \rightarrow F_{net} = ma = m(0) = 0$.

Disagree: $F_{net} = 0$, because $F_D = F_g$

Newton's first law: an object moving at constant velocity has zero net force on it. A net force changes the velocity; no force is needed to keep it moving.

Watch out: the velocity is down, but the net force follows the acceleration, and the acceleration is zero.

2. Rocket burning fuel (25 points)

Step 1 **Up is +. List what is given.**

M (liftoff)	F_{th}	a (later)	m (later)
3.40×10^4 kg	5.10×10^5 N, up, constant	+7.2 m/s²	?

(a) rocket, at liftoff



Only two forces: thrust F_{th} up (the exhaust pushing on the rocket) and gravity F_g down. The rocket speeds up from rest, so F_{th} is longer.

Step 2 **(a) Newton's second law at liftoff.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_{th} - F_g}{M} \rightarrow a = \frac{F_{th} - Mg}{M} \rightarrow a = \frac{F_{th}}{M} - g$$

$$a = \frac{5.10 \times 10^5 \text{ N}}{3.40 \times 10^4 \text{ kg}} - 9.8 \text{ m/s}^2 = 15.0 \text{ m/s}^2 - 9.8 \text{ m/s}^2 = 5.2 \text{ m/s}^2$$

$a = 5.2$ m/s², upward (1 N/kg = 1 m/s²)

Step 3 **(b) Same law, but now the mass m is the unknown.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_{th} - mg}{m} \rightarrow ma = F_{th} - mg \rightarrow ma + mg = F_{th} \rightarrow m(g + a) = F_{th} \rightarrow m = \frac{F_{th}}{g + a}$$

$$m = \frac{5.10 \times 10^5 \text{ N}}{9.8 \text{ m/s}^2 + 7.2 \text{ m/s}^2} = \frac{5.10 \times 10^5 \text{ N}}{17.0 \text{ m/s}^2} = 3.00 \times 10^4 \text{ kg} \rightarrow$$ **$m = 3.00 \times 10^4$ kg**

Three significant figures, matching 5.10×10^5 N and 17.0 m/s²; 3.0×10^4 kg also earns credit.

Step 4 **(b) Fuel burned: the mass that is gone.**

$$\Delta m = M - m = M - \frac{F_{th}}{g + a} = 3.40 \times 10^4 \text{ kg} - 3.00 \times 10^4 \text{ kg} \rightarrow$$ **4.0×10^3 kg of fuel**

Watch out: use the mass at that moment (m) in both mg and ma, not the liftoff mass M.

Step 5 **(c) Why the acceleration keeps growing.**

From (a): $a = \frac{F_{th}}{m} - g$. F_{th} and g stay the same, and burning fuel makes m smaller, so $\frac{F_{th}}{m}$ gets bigger and a gets bigger.

In force terms: the weight mg shrinks, so $F_{net} = F_{th} - mg$ grows, and that bigger net force acts on a smaller mass. **smaller m: same thrust, smaller weight, bigger a**

Step 6 **(d) Third law last.**

$F_{\text{exhaust on rocket}} = -F_{\text{rocket on exhaust}}$. The thrust is the exhaust pushing up on the rocket, so its partner is the rocket pushing down on the exhaust gases.

$F_{\text{rocket on exhaust}} = 5.10 \times 10^5$ N, downward

It acts on the exhaust, a different object, so it never goes on the rocket's diagram.

3. Elevator going up (25 points)

Step 1 **Up is +. Solve in letters once.**

$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_N - F_g}{m} \rightarrow F_N - mg = ma \rightarrow F_N = m(g + a)$

(a) person, speeding up



Step 2 **(a) Moving up and speeding up: $a = +1.4$ m/s².**

The acceleration is up, so F_{net} is up: the F_N arrow is longer than F_g.

$F_N = (62\text{ kg})(9.8\text{ m/s}^2 + 1.4\text{ m/s}^2) = (62\text{ kg})(11.2\text{ m/s}^2) = 694.4$ N

$F_N = 6.9 \times 10^2$ N, upward (two significant figures, matching 62 kg)

Step 3 **(b) Constant speed: $a = 0$.**

$F_N = m(g + 0) = mg = (62\text{ kg})(9.8\text{ m/s}^2) = 607.6$ N

$F_N = 6.1 \times 10^2$ N, upward

Equal arrows: the floor pushes up with exactly the person's weight.

(c) person, slowing down



Step 4 **(c) Moving up but slowing down: the acceleration points down, $a = -2.1$ m/s².**

$F_N = m(g + a) = (62\text{ kg})(9.8\text{ m/s}^2 + (-2.1\text{ m/s}^2))$

$= (62\text{ kg})(7.7\text{ m/s}^2) = 477.4$ N

$F_N = 4.8 \times 10^2$ N, upward

F_N is now shorter than F_g, so the net force points down.

Watch out: the elevator is still moving up in (c). The sign of a comes from slowing down, not from the direction of motion.

Step 5 **(d) Third law last.**

$F_{\text{person on floor}} = -F_{\text{floor on person}}$: same size as F_N from (a), opposite direction.

6.9×10^2 N, downward on the floor (Newton's third law)

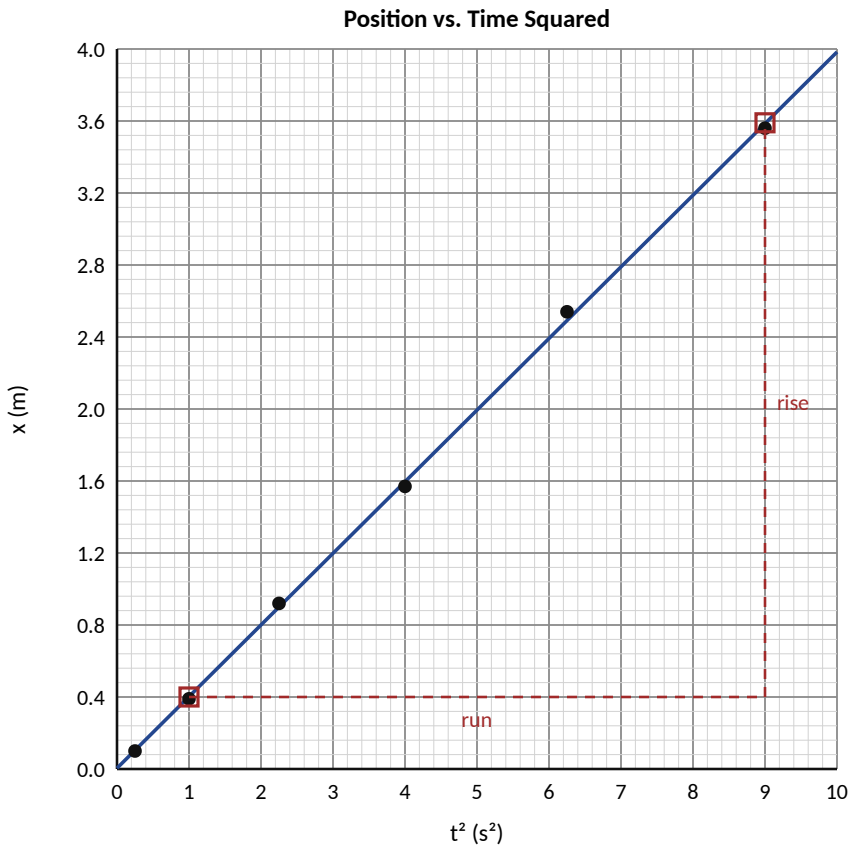
4. Linearize the data (25 points)

Step 1 **(a) Square each time; units square too.**

$t^2 = (0.50\ s)^2 = 0.25\ s^2$

t (s)	x (m)	t^2 (s^2)
0.50	0.10	0.25
1.00	0.39	1.00
1.50	0.92	2.25
2.00	1.57	4.00
2.50	2.54	6.25
3.00	3.56	9.00

Step 2 **(b) Graph: x on the vertical axis, t^2 on the horizontal axis, labeled with units, a scale that uses more than half the grid.**



One straight best-fit line through the middle of the points. The red squares are points read off the line, not data points.

Step 3 **(c) Slope from two far-apart points on the line, units carried.**

Points on the line: (1.00 s^2, 0.40 m) and (9.00 s^2, 3.59 m)

$$\text{slope} = \frac{3.59\ \text{m} - 0.40\ \text{m}}{9.00\ \text{s}^2 - 1.00\ \text{s}^2} = \frac{3.19\ \text{m}}{8.00\ \text{s}^2} = 0.399\ \text{m/s}^2$$

slope ≈ 0.40 m/s^2 (m ÷ s^2 = m/s^2)

Any value from 0.37 m/s^2 to 0.43 m/s^2 from a reasonable line earns credit.

Step 4 **(d) What the slope means: match $x = \frac{1}{2}at^2$ to $y = mx + b$.**

$x = \frac{1}{2}a \cdot t^2 + 0$

vertical (y)	horizontal (x)	slope	intercept
x (m)	t^2 (s^2)	½a	0 (starts from rest at x = 0)

the slope is ½a, half the acceleration

Step 5 **(d) Acceleration, then the net force.**

slope = ½a → a = 2 × slope = 2(0.399 m/s^2) = 0.7975 m/s^2 → **a ≈ 0.80 m/s^2**

$a = \frac{F_{net}}{m}$ → F_{net} = ma = (2.0 kg)(0.80 m/s^2) = 1.60 N → **F_{net} ≈ 1.6 N**

Accepted from a reasonable line: a from 0.74 m/s^2 to 0.86 m/s^2, and F_{net} from 1.5 N to 1.7 N.

Watch out: the slope is ½a, not a. Multiply by 2 before multiplying by the mass.

1. Free-body diagrams

Suggested partial credit	pts
(a) Thrust forward and drag backward, thrust longer	4
(a) Lift up and normal force up as two separate arrows, gravity down; every force labeled	4
(a) The two up arrows together as long as gravity; no friction arrow	2
(b) Drag up and gravity down, equal lengths, labeled	5
(c) Less than the weight	2
(c) $F_N + F_L = F_g$, so $F_N = F_g - F_L$ (lift carries part of the weight)	3
(d) Disagree: terminal speed is a constant velocity, so a = 0	3
(d) $F_{net} = 0$ ($F_D = F_g$); motion does not need a net force (first law)	2
Total	**25**

2. Rocket burning fuel

Suggested partial credit	pts
(a) Thrust up, longer than gravity down; labeled; no other forces	3
(a) $F_{th} - Mg = Ma$ solved for a in letters	3
(a) 5.2 m/s², upward	2
(b) $m = F_{th}/(g + a)$ in letters	3
(b) 3.00×10^4 kg (3.0×10^4 kg accepted), units in every parenthesis	2
(b) Fuel burned = $M - m = 4.0 \times 10^3$ kg	3
(c) Thrust is constant and the mass decreases as fuel burns	2
(c) $a = F_{th}/m - g$: smaller m gives larger a (smaller weight, bigger net force)	3
(d) The rocket pushes on the exhaust gases	2
(d) 5.10×10^5 N, downward	2
Total	**25**

3. Elevator going up

Suggested partial credit	pts
(a) Normal force up, longer than gravity down; labeled	3
(a) $F_N - mg = ma$ solved for F_N in letters	3
(a) 6.9×10^2 N, upward, units in every parenthesis	3
(b) a = 0, so $F_N = mg$	3
(b) 6.1×10^2 N	2
(c) $a = -2.1$ m/s² (slowing while moving up means the acceleration is down)	4
(c) 4.8×10^2 N	2
(d) The person pushes on the floor (third law)	2
(d) 6.9×10^2 N, downward	3
Total	**25**

4. Linearize the data

Suggested partial credit	pts
(a) t^2 column correct, units s^2	4
(b) x on the vertical axis, t^2 on the horizontal axis	2
(b) Axes labeled with quantity and units	2
(b) Scale uses more than half the grid; points plotted correctly	3
(c) One straight best-fit line through the middle of the points	2
(c) Two far-apart points on the line, not data points	3
(c) Slope between 0.37 m/s² and 0.43 m/s², with units	2
(d) Slope = ½a, from $x = \frac{1}{2}at^2$	3
(d) a = 2 × slope, between 0.74 m/s² and 0.86 m/s²	2
(d) $F_{net} = ma$, between 1.5 N and 1.7 N	2
Total	**25**